

Postulates, Theorems, Converses and Biconditionals

High School Geometry Definitions, Postulates, and Proof Logic

Classifying postulates and theorems, applying a postulate, testing the truth of a converse, and deciding when a biconditional is valid

Four words, four jobs. A **definition** says what a term means and is reversible by agreement. A **postulate** is accepted without proof, so the system has somewhere to start. A **theorem** is proved from definitions, postulates and earlier theorems — you can ask “prove it” and get an answer. A **biconditional** (“ p if and only if q ”) packs a statement and its converse into one sentence, and is valid only when *both* are true. Obviousness decides nothing; provability does.

Directions. Name the statement that authorises each equation you write. A converse is never true automatically: disprove one with a single counterexample, or say what proves it. Write the requested value on the answer line.

1. B lies between A and C , with $AB = 2x + 3$, $BC = 3x - 1$, and $AC = 27$. **(a)** Name the statement that permits $AB + BC = AC$, and say whether it is a definition, a postulate or a theorem. **(b)** Solve for x .

Answer: _____

2. Classify each statement as a **definition**, a **postulate** or a **theorem**, with the one-phrase reason that decides it.

Statement	Type	Reason
Through any two points there is exactly one line.		
An angle bisector divides an angle into two congruent angles.		
Vertical angles are congruent.		
If M is the midpoint of $\overline{AB}$, then $AM = MB$.		
The angles of a triangle sum to 180° .		

3. $\overrightarrow{BD}$ lies in the interior of $\angle ABC$, with $m\angle ABD = 4x$, $m\angle DBC = 3x + 5$ and $m\angle ABC = 96^\circ$. (a) Name the postulate that lets you add the two smaller measures. (b) Solve for x , then check that your two measures add to 96.

Answer: _____

4. Consider the true statement: *if $x = 6$, then $x^2 = 36$* . **(a)** Write its converse. **(b)** Solve $x^2 = 36$ completely and list *every* real solution on the answer line. **(c)** Use the solution set to decide whether the converse is true, naming the counterexample if it is not.

Answer: _____

5. Consider the biconditional: $2x - 7 = 9$ *if and only if* $x = 8$. **(a)** Write the two conditionals it packs together. **(b)** Solve $2x - 7 = 9$ and give the complete solution set. **(c)** Explain why the solution set being *exactly* one value is what makes the biconditional valid.

Answer: _____

6. The Vertical Angles Theorem: if two angles are vertical angles, then they are congruent. Now let $\angle 1$ and $\angle 2$ be *adjacent* angles (they share a side and are not vertical) with $m\angle 1 = 5x - 8$, $m\angle 2 = 3x + 22$, and $\angle 1 \cong \angle 2$. **(a)** Write the converse of the theorem. **(b)** Solve for x and find both measures. **(c)** Explain how this pair settles the truth of the converse.

Answer: _____

7. “Through any two points there is exactly one line” is a **postulate**. “If two distinct lines intersect, then they intersect in exactly one point” is a **theorem**. Both sound equally obvious. Explain what makes the second a theorem while the first is a postulate, and name the statement a proof of the second would rest on.

8. A useful biconditional: *a point P lies on the perpendicular bisector of $\overline{AB}$ if and only if $PA = PB$* . For a point P in the plane, $PA = 4x - 5$ and $PB = 2x + 7$. **(a)** Solve for the x that places P on the perpendicular bisector. **(b)** Find PA and PB and check they agree. **(c)** Which half of the biconditional did you use — the half starting from equal distances, or the half starting from the bisector? Say why.

Answer: _____

9. Consider the true statement: *if a quadrilateral is a square, then it has four right angles*.
(a) Write its converse. (b) Decide whether the converse is true. (c) If it is false, sketch and label a quadrilateral satisfying the hypothesis but not the conclusion, and name the property of a square your figure fails. (d) Say what this means for the statement “a quadrilateral is a square if and only if it has four right angles”.

10. (a) A definition is always reversible. Rewrite as a single biconditional: *an angle bisector is a ray that divides an angle into two congruent angles*. (b) Now take: *if two angles are complementary, then they are adjacent*. Can it be rewritten as a valid biconditional? Justify by examining the conditional and its converse separately, and give a counterexample for whichever direction fails.